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### Abstract

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| Title | Self-localization of indoor scenes using a simply 3-D model with wireframe and omnidirectional image textures |
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Recently, as digital twin technologies become widespread, expectations for localization and navigation technologies in indoor spaces are rising. However, the high introduction cost of dedicated equipment such as 3-D measurement devices and beacons, and the instability of Visual SLAM in indoor environments with featureless walls, are impeding widespread adoption. Especially in navigation applications, operation stability in featureless environments and cost reduction are more important than geometrically rigorous shape reconstruction.

Therefore, in this study, we propose a practical self-localization method that uses only existing building information such as 2-D maps and omnidirectional images. We construct a 3-D model of the indoor environment as a wireframe model that simply represents the environment, and it is generated by assigning images captured with an omnidirectional camera as textures. In the making of the model, we propose a method in which the user specifies small number of the feature points that become vertices of the wireframe on the 2-D map. The remaining vertex alignment, mesh generation, and texture assignment are performed automatically. Specifically, the wireframe vertices are determined by the camera positions, and the mesh is divided accordingly, and images are generated with the camera's optical axis facing each mesh centroid, then assigned as textures. Furthermore, if the angle between the mesh normal and the camera's optical axis is large, we apply texture blend processing to images from multiple viewpoints improves visual quality. To correspond with the generated model with the input images, we use SuperPoint and SuperGlue, which are deep learning-based methods, for feature detection and matching. We compared this method with the method of SIFT and AKAZE and confirmed that our proposed approach is significantly robust to featureless scenes and large viewpoint changes and achieved higher matching success rate. In addition, self-localization using the PnPL method proposed in our laboratory achieved valid estimation in approximately 95% of successfully matched frames. We also integrated this method into an indoor navigation system and conducted a demonstration experiment. By combining it with a mobile device's VIO (Visual-Inertial Odometry), we achieved real-time performance and stable estimation. As a result, smooth and high-accurate navigation was demonstrated even in environments with few detectable feature points, with VIO ensuring motion continuity while the proposed method corrects accumulated errors.